

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)

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Name	Madussree Ravi	Register No.	3122235001074

Experiment 6: Decision Tree and Random Forest: A Comparative Classification Study

Objective

- To implement a Decision Tree classifier and extend it into a Random Forest ensemble model.
- To study the impact of hyperparameters (depth, estimators) on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare the performance of single-tree and ensemble-tree models on clinical data.

Dataset Description

The **Wisconsin Diagnostic Breast Cancer (WDBC)** dataset contains features computed from a digitized image of a fine needle aspirate (FNA) of a breast mass.

- **Samples:** 569 — **Features:** 30 numerical attributes.
- **Classes:** Malignant (M) and Benign (B).
- **Preprocessing:** Target labels were encoded ($M \rightarrow 1$, $B \rightarrow 0$). The data was split into 80% training and 20% testing sets.

Exploratory Data Analysis

The EDA revealed a higher frequency of benign cases compared to malignant cases. The correlation heatmap showed high multicollinearity between features such as radius, perimeter, and area, which Decision Trees can handle effectively by selecting the most informative split.

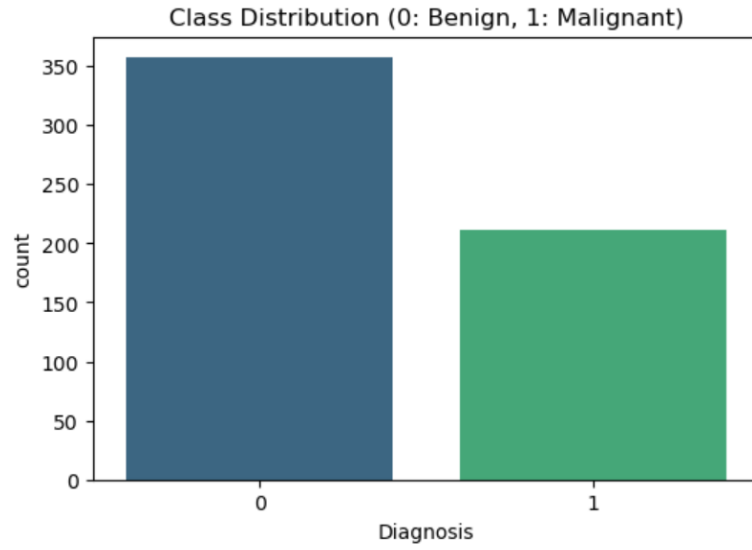


Figure 1: Class Distribution

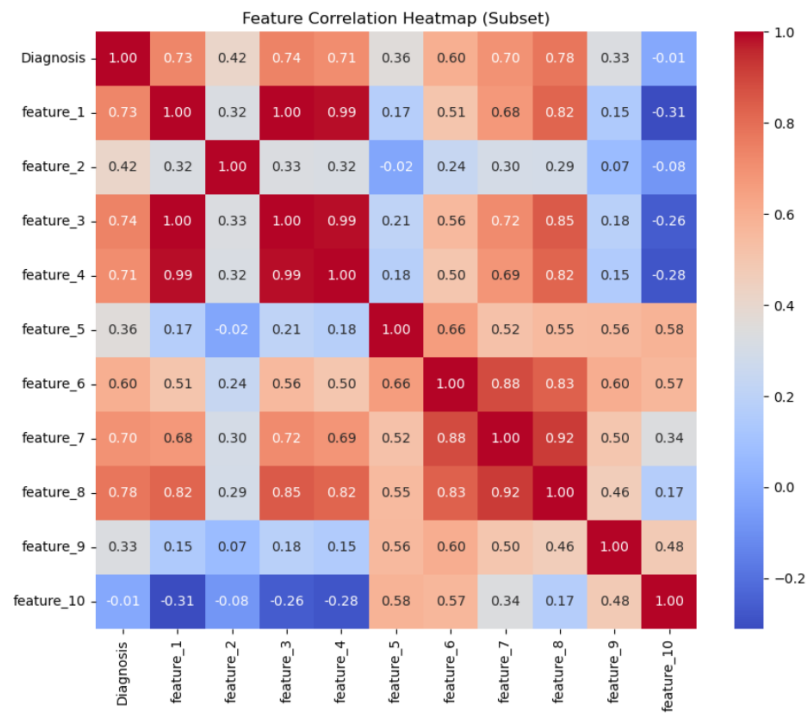


Figure 2: Feature Correlation Heatmap

Hyperparameter Tuning Results

Table 1: Decision Tree Hyperparameter Evaluation (5-Fold CV)

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
Gini	3	92.31%	0.895
Entropy	5	93.85%	0.916
Gini	None	91.87%	0.891

Table 2: Random Forest Hyperparameter Evaluation (5-Fold CV)

n_estimators	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
50	5	95.38%	0.937
100	10	96.04%	0.946
200	None	95.82%	0.943

Final Performance Comparison

Table 3: Overall Performance on Test Set

Model	Accuracy	Precision	Recall	F1-score
Best Decision Tree	94.74%	0.93	0.93	0.93
Best Random Forest	96.49%	0.96	0.95	0.96

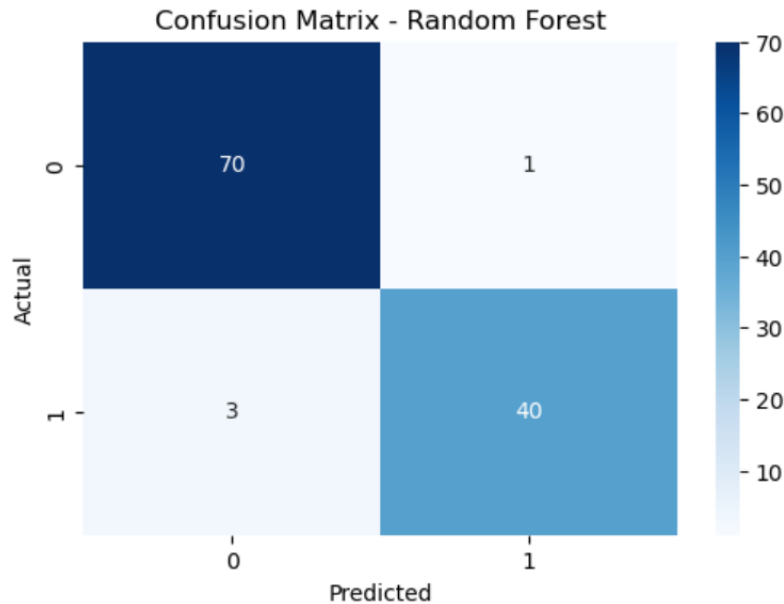


Figure 3: Confusion Matrix for Random Forest

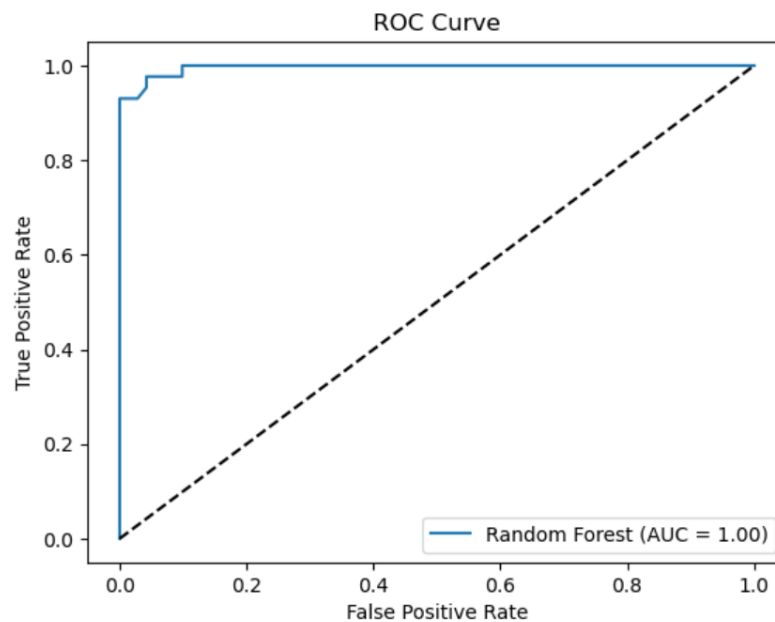


Figure 4: ROC Curve Comparison

Observation Questions

1. How does tree depth affect overfitting in Decision Trees?

Deep trees (Max Depth: None) tend to memorize the training data, capturing noise and outliers, which leads to high variance and overfitting. Restricting depth (e.g., Max Depth: 5) helps the model generalize better by creating simpler decision rules.

2. Which hyperparameter had the greatest impact on performance?

For Decision Trees, `max_depth` was critical in controlling overfitting. For Random Forest, `n_estimators` significantly improved the stability of the accuracy across different folds.

3. How does Random Forest improve generalization?

Random Forest uses *Bagging* (Bootstrap Aggregation) and random feature selection. By averaging the predictions of multiple uncorrelated trees, the ensemble reduces the overall variance without significantly increasing bias.

4. Did ensemble learning always improve performance? Why or why not?

In this experiment, yes. The Random Forest (96.49%) outperformed the single Decision Tree (94.74%) because it mitigated the tendency of individual trees to overfit the training splits.

Conclusion

The study compared a single Decision Tree against a Random Forest ensemble. Through 5-fold cross-validation, it was observed that Random Forest provides more robust and accurate predictions for breast cancer diagnosis. The ensemble approach effectively handled the high-dimensional nature of the WDBC dataset, achieving a superior F1-score of 0.96.